

Stephen Marn

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Full-stack data engineer in manufacturing, owning the lifecycle from sensor and IoT ingestion through pipeline automation, warehousing, and analytics. Designed a hybrid data platform projected to save \$850K+ over five years, recovered production-critical systems, and unlocked previously inaccessible data in high-uptime plant environments.

EXPERIENCE

Data Analyst / Engineer | Banknote

January 2025 - Present

Security printing manufacturer (currency and secure documents)

- Unlocked two years of trapped production data for 48 machines, enabling every machine performance KPI leadership now tracks. Python ETL pipeline parsing daily PDF reports into validated SQL Server tables.
- Cut recurring report delivery from days to minutes, recovering an estimated **\$18K to \$37K** per year in analyst capacity. Automated pipelines replacing manual assembly, extraction, and quality reporting, eliminating permanent data loss risk.
- Designed the company's end-to-end data platform, projected to save **\$850K+** over five years. SQL Server 2025, Dagster, Grafana, and Power BI Report Server on a cloud-ready medallion architecture, so future migration requires no rework.
- Recovered a production-critical quality inspection system from a multi-week outage and have run it solo since. **30M+** inspection records across three presses, restored in-house with no vendor support.
- Coordinate factory digital twin development and a company-wide digital waste tracking system. Advise R&D on data platform capabilities and constraints.
- Delivered enterprise BI at zero license cost, serving **15+** users across **20+** dashboards. Scaled Grafana into a governed platform with access controls and standard conventions, establishing the company's first reporting-layer data governance.
- Deliver forecasting and capacity planning analytics used in production scheduling. Descriptive and predictive models for equipment effectiveness and demand, fed by newly automated data sources.

Data Analyst | nVent

May 2023 - November 2024

Global manufacturer of electrical connection and protection solutions

- Replaced 3-hour data delays with real-time reporting for **30+** daily users in Finance and Operations. Built and maintained live data pipelines across enterprise systems.
- Cut manual reporting work **50%** across recurring deliverables. Automated SQL reporting processes end to end.
- Ensured consistent, accurate reporting across enterprise systems. Mapped data flows spanning ERP and Snowflake, prioritizing report and tool development with operations, engineering, and finance stakeholders.
- Earned Performance Accountability Award for measurable business impact through analytics.

Anti-Money Laundering Analyst | AML RightSource

March 2021 - May 2023

Managed services provider for financial crime compliance

- Cut manual processing time **50%** in customer onboarding. Analysis-driven workflow improvements adopted as standard practice.
- Eliminated a **3,000+** application backlog in 3 months. Data-driven process optimization and workflow standardization.
- Flagged suspicious transaction activity across fintech and digital payment platforms. Statistical anomaly detection methods and analyst tooling, translating regulatory requirements into technical specifications.

Air Defense Officer | Army National Guard

May 2020 - Present

- Lead teams of 4 to 80 operating **\$5M+** in networked radar, sensor, and command and control systems. Integrated real-time data across platforms in 5 major exercises and NATO operations in Latvia and Lithuania.

FEATURED PROJECTS (public, verifiable: code, CI history, and live dashboards)

Manufacturing PDF-to-Warehouse Pipeline · github.com/Steve27M/manufacturing-pdf-etl

Public, synthetic-data mirror of real plant reporting problems: daily PDF reports into a tested analytics warehouse with a live dashboard. pdfplumber, Dagster, and dbt medallion into a DuckDB Kimball star; bad rows quarantined with lineage instead of silently dropped.

Live Flight Telemetry Pipeline · github.com/Steve27M/opensky-flight-pipeline

Warehouses live aircraft telemetry from a rate-limited, always-moving API source, the same ingestion problems sensor fleets create. Incremental polling with retry handling, append-only landing with lineage, dbt Kimball star, live geospatial map.

HelioSentinel: Space Weather Pipeline · github.com/Steve27M/heliosentinel

Detects dangerous space weather from live NOAA and NASA feeds and drafts plain-English incident reports, graded against NASA's event log. Deterministic SQL detection kept separate from LLM narration; schema-validated output with quarantine; prompt changes CI-gated (severity accuracy **37%** to **83%**).

MLB Analytics Pipeline · github.com/Steve27M/mlb-analytics

Rebuilds **2.18M** pitch records into 13 statistical models and a live dashboard nightly, with CI refusing to ship on any failed check. DuckDB and dbt (Kimball, SCD2); every model implemented in both R and Python behind a parity gate; win model evaluated on a sealed holdout.

EDUCATION

Georgia Institute of Technology, M.S. Analytics (Computational Analytics) · Machine learning, predictive modeling, data engineering Expected 2029

Mercyhurst University, M.S. Applied Intelligence · GPA 3.9; Capstone: statistical methodology for assessing foreign mis/disinformation campaigns; Coursework: competitive intelligence, strategic intelligence, intelligence research methods 2024

Virginia Military Institute, B.A. International Studies · Army ROTC; Cadre (cadet leadership) 2020

CERTIFICATIONS

AWS Certified Cloud Practitioner 2026

MIT Applied Data Science Program 2023

SKILLS

Languages: Python, SQL, R

Data Platform: SQL Server, Snowflake, DuckDB, dbt, Dagster, ETL/ELT pipeline design, Lakehouse / medallion architecture, Kimball dimensional modeling, CI for data (GitHub Actions)

BI & Observability: Power BI, Grafana, Prometheus, SAP Analytics Cloud

Cloud: AWS (hybrid architecture, migration planning)

ML & Statistics: scikit-learn, XGBoost, Statistical modeling, Model evaluation and holdout design, Forecasting

Manufacturing: IoT telemetry, OEE analytics, Digital twin coordination

Working knowledge: C++ (software engineering coursework), TensorFlow / PyTorch (MIT ADSP coursework)